GUESS for Text, Knowledge and Citation Networks

Character and word networks from text, citation analysis and main paths, co-authorship, keyword maps, document similarity and knowledge graphs

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Why networks?

Texts and the literature about them are full of implicit networks. Characters share scenes, words share sentences, papers cite papers, authors write together, and documents share vocabulary. Making these networks explicit answers questions such as:

- Who is at the centre of a story, and who connects its worlds?
- What are the themes of a corpus, and how do they relate?
- Where did a research field come from, and how is it organised?

This book assumes the Kitchen for loading, querying and drawing. Its data are in the cookbooks' `data` folder:

file	what it is
`scandal-in-bohemia.txt`	Arthur Conan Doyle's "A Scandal in Bohemia" (1891), the first Sherlock Holmes short story. Public domain, from Project Gutenberg.
`vis-papers.gdf`	1,044 highly cited papers whose titles mention graph drawing, graph layout or network/graph visualisation (1980–2024), with the 4,073 citations among them (citing → cited). Node fields: `title`, `year`, `cites` (citations from anywhere), `firstauthor`, `venue`. From OpenAlex, CC0.
`vis-papers-nodes.csv`	the same papers as a node table
`vis-paper-authors.csv`	paper, author: one row per authorship
`vis-paper-keywords.csv`	paper, keyword, score: OpenAlex's keywords for each paper
`polblogs-sample.gdf`	600 US political blogs from 2004 and the 2,786 links among them (a random sample of Adamic and Glance's 1,490), directed. Node fields: `label` (the blog's address), `leaning` (liberal or conservative). From Mark Newman's network data page.
`vis-institutions.gdf`	the 932 institutions of those papers' authors, located (`latitude`, `longitude`, `country`), with `papers` (how many) and `firstyear` (their first paper in the set), linked when they co-author (edge `firstyear`: their first joint paper). From OpenAlex, CC0.

Networks from text

Map the characters of a story Basic

You have a story. **You want** its cast as a network: who appears with whom.

Ingredients: `scandal-in-bohemia.txt` · `makeFromText`

```
cast = {"Holmes": ["Holmes", "Sherlock"], "Watson": ["Watson", "Doctor"],
        "Irene Adler": ["Irene Adler", "Irene", "Miss Adler", "the woman", "Mrs.
         ↪ Norton"],
```

```
        "the King": ["King", "Majesty", "Count Von Kramm", "Von Kramm"],
        "Godfrey Norton": ["Godfrey Norton", "Godfrey", "Mr. Norton"],
        "John": ["John"], "Mrs. Turner": ["Mrs. Turner"]}
makeFromText("scandal-in-bohemia.txt", cast)
sizeBy(count, 10, 50)
sizeBy(weight, 1, 10)
labelBy(label)
circleLayout()
```

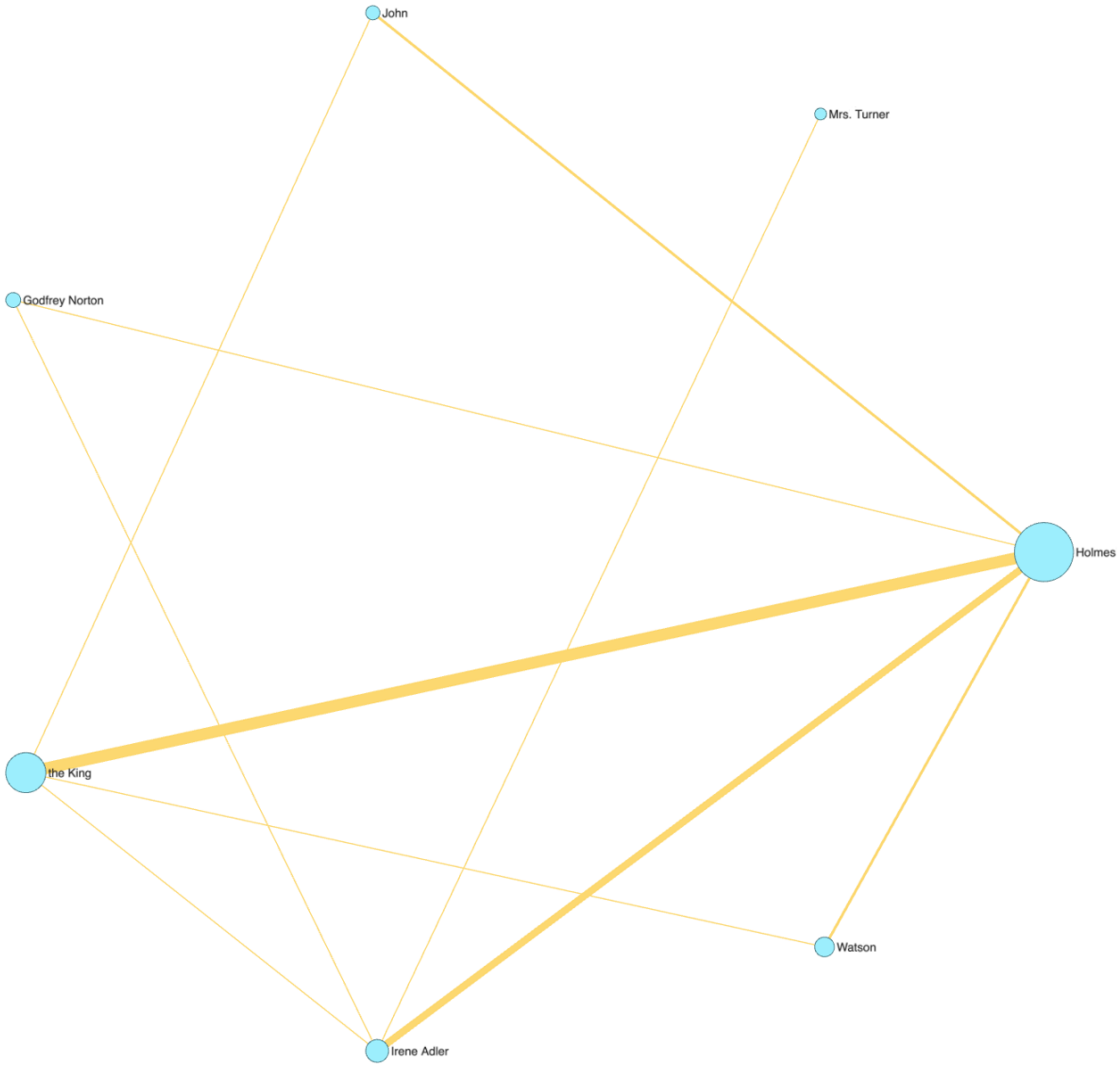


Figure 1: Who shares paragraphs with whom in “A Scandal in Bohemia”

How it works. `makeFromText(file, terms)` splits the text into paragraphs and links two characters when they're named in the same one. `weight` counts those paragraphs. Each character is a key in the dictionary, and its list holds every way the text refers to them, so "Irene Adler", "Miss Adler" and "the woman" all count for one node. Nodes get `count` (mentions), `units` (paragraphs) and `first` (the first paragraph they appear in). The strongest tie is Holmes and the King (8 paragraphs), then Holmes and Irene Adler (5). The story is about the contest between them, conducted through a client.

Variations. `by="sentence"` uses smaller units, giving fewer, more intimate links. `window=15` links names within 15 words of each other. `scatterLayout(first, count)` puts each character at the point where they enter the story, which gives a rough cast timeline.

Pitfalls. Mentions aren't presence. Watson is named only 11 times because he narrates, as "I". A first-person narrator is everywhere and nowhere in a co-occurrence network. Pronouns ("he", "she") aren't counted at all. Treat the network as "who is named together".

Build a word network Intermediate

You have text with no list of names. **You want** its main words and how they cluster.

```
makeFromText("scandal-in-bohemia.txt", maxTerms=50, minCount=4,
             stopwords=["cried", "come", "came", "asked", "looked", "know",
 ↪  "see"])
remove(weight < 2)
louvainClusters()
colorize(louvain_class)
sizeBy(count, 6, 30)
labelBy(label)
forceAtlas2Layout()
```

How it works. Without `terms`, `makeFromText` takes the `maxTerms` most frequent words that occur at least `minCount` times, ignoring a built-in list of common English words (*the*, *of*, *said*…) and any `stopwords` you add. `remove(weight < 2)` drops pairs that co-occur only once, the usual first step against a hairball. Communities then group words that keep company.

Pitfalls. In a single short story, frequent words are mostly the story's furniture (*room*, *door*, *street*), and communities are weak (modularity around 0.2). Word networks earn their keep on larger corpora (many abstracts, many speeches), with a domain stopword list and a stricter weight threshold.

Citation networks

Explore a citation network Basic

Ingredients: `vis-papers.gdf` · `indegree` · `pagerank` · `generation`

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
for n in sortBy(indegree)[-5:]: print(n.indegree, n.cites, n.label, n.title[:50])
for n in sortBy(pagerank)[-5:]: print(round(n.pagerank, 3), n.label,
 ↪  n.title[:50])
```

How it works. Edges run from the citing paper to the cited one, so `indegree` counts citations *within this set*. `cites` is the paper's citation count from anywhere, which is much larger. Fruchterman and Reingold's 1991 force-directed paper is the most cited here. `pagerank` rewards papers cited by well-cited papers, so it favours foundations: the annotated bibliography of Di Battista et al. (1994) and Eades's 1984 spring heuristic rank highest. `generation` is each paper's topological level, counting back through chains of citation. `removeDisconnected()` drops the 265 papers with no citation links to the rest of the set.

Cumulative advantage Intermediate

You want to know why a few papers collect most of the citations, and whether "the cited get cited more" is enough to explain it.

Ingredients: `makePriceModel` · `born` · `revealUpTo` · `fitPowerLaw`

```
makePriceModel(500, c=3, a=1, seed=1)
sizeBy(indegree, 4, 30)
colorBy(born, "200,60,40", "120,160,230")
center(g.nodes)
revealUpTo(150)
```

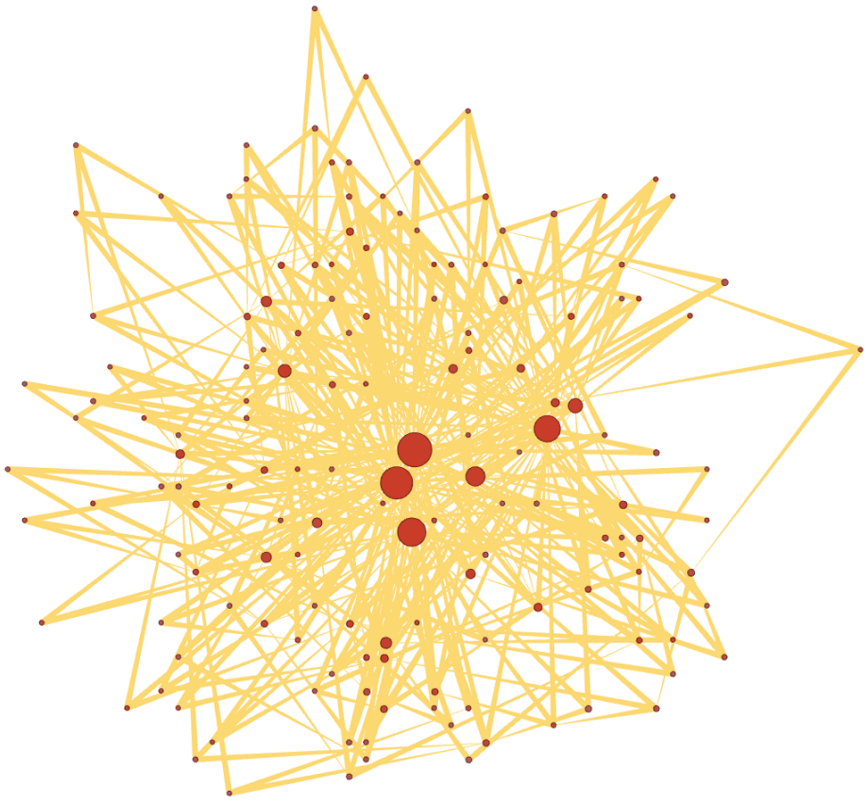


Figure 2: Price's model after 150 of 500 papers: early papers (red) already collect the citations

How it works. In Price's model (1976) papers arrive one at a time and each cites `c` earlier ones, chosen in proportion to their citations so far plus a head start `a`: *cumulative advantage*. `makePrice-Model(n, c, a)` grows such a network and records when each paper (and citation) arrived in the field `born`. `revealUpTo(step)` shows the network as it was at that step, with every paper already in its final place, so calling it with growing steps (or from a timer) replays the growth. The oldest papers end up the most cited: they have had the longest to accumulate.

The model predicts a power-law tail with exponent $\alpha = 2 + a/c$, and a power law in $k + a$ rather than in k, so fit it with `shift=a`:

```
makePriceModel(20000, c=3, a=1, seed=1, layout=false)
fitPowerLaw(indegree, shift=1)            # alpha about 2.24; theory 2.33
makeFromGDF("vis-papers.gdf")
fitPowerLaw(indegree)                     # the real set: about 3.0, from 13
 ↪  citations up
```

The real graph-drawing papers have a steeper tail ($\alpha \approx 3.0$ from 13 citations up, only 80 papers in the tail) than the model with the same mean of about 4 references per paper. A citation set chosen by topic is a slice of the literature, not a closed system.

Pitfalls. Fitted exponents come out below the formula on finite networks, and creep up as n grows (2.09 at 2,000 papers, 2.24 at 20,000). Fitting `indegree` without `shift=a` reads lower still. A good fit is not evidence of the mechanism: other models, such as copying citations from a paper you read, give the same tail.

What is this collection about? Basic

You want a quick look at what a set of documents covers, and anything that doesn't belong.

Ingredients: `vis-papers.gdf` · `wordCloud`

```
makeFromGDF("vis-papers.gdf")
wordCloud(title, top=60)
```

How it works. `wordCloud(field)` counts the words of a text field, leaves out common English words, and draws the most frequent ones sized by count. It is linked like the other charts: hover a word to light up the papers whose title contains it, and click it to select them.

Most of the words are what you'd expect from graph drawing, such as "graph", "visualization", "layout" and "force". But "forcing", "radiative" and "aerosol" are there too. Ninety of the papers are climate science about the "direct radiative forcing" of aerosols: the OpenAlex title search for "force-directed" that helped collect the set matched them too. Hovering "radiative" shows they form an island of their own: 247 citations among themselves and none to or from the rest. Leave out the words you've already understood to see what comes next:

```
makeFromGDF("vis-papers.gdf")
wordCloud(title, stopwords=["graph", "graphs", "visualization", "drawing",
 ↪  "network", "networks"])
offtopic = [n for n in g.nodes if "forcing" in n.title.lower()]
len(offtopic)
```

Pitfalls. Counts are of words, not ideas: "network" and "networks" are separate, and short phrases such as "force-directed" break into two words.

Find the main path of a field Intermediate

You want the backbone of a field's history: the chain of papers through which most of its citation paths run.

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
```

```
mp = mainPath()
showOnly(mp)
scatterLayout(year, generation)
mp.color = red
labelBy(label)
g.nodes.labelpos = "outside"
optimizeLabels()
```

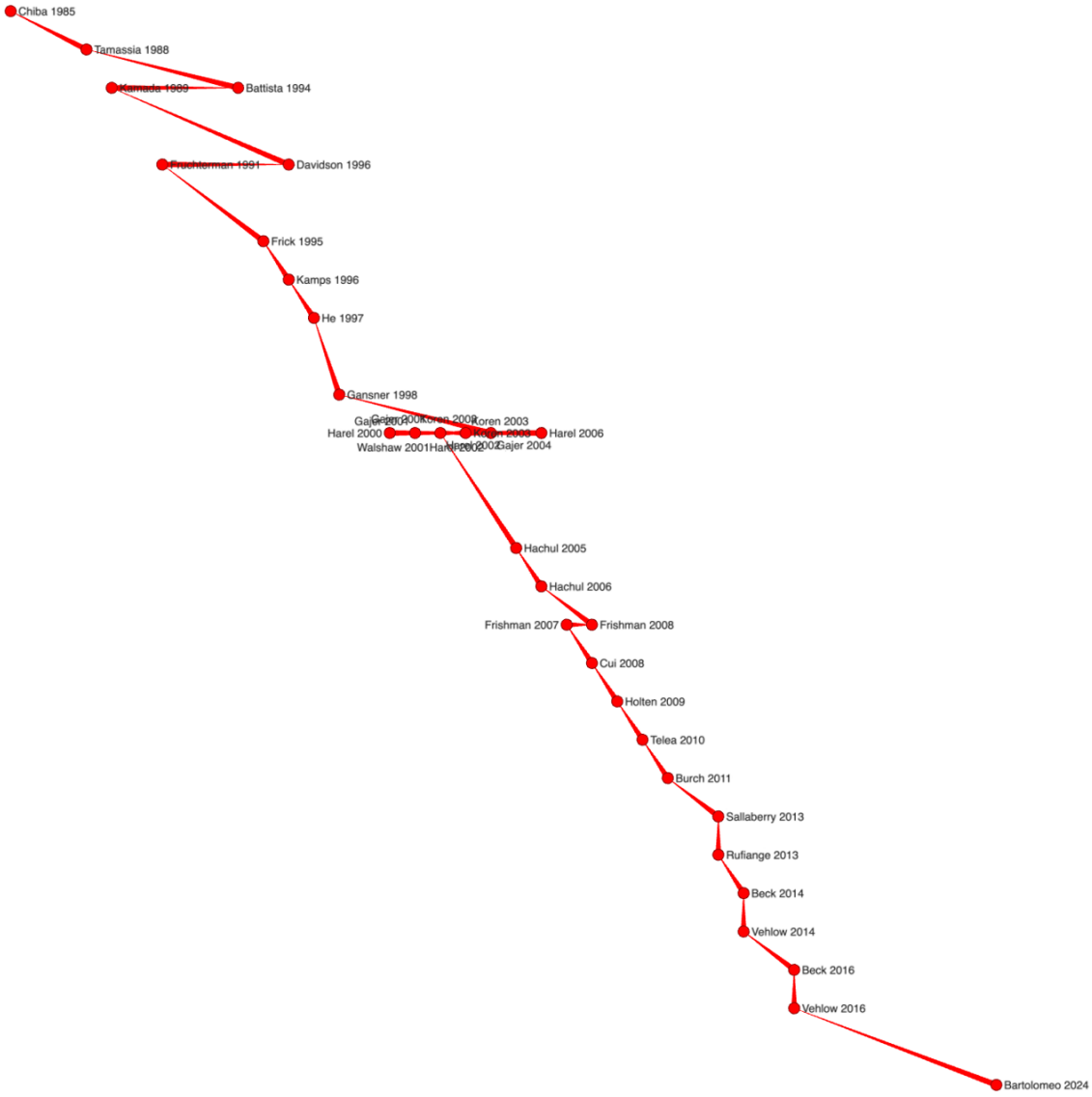


Figure 3: The main path of graph-drawing research

How it works. `mainPath()` performs main path analysis (Hummon and Doreian). For every cita-

tion it counts how many chains from the newest papers back to the oldest pass through it (its *search path count*, stored as `spc`, a share of all such chains). Then it follows the heaviest links. `mp` holds the path's papers and citations, and `showOnly(mp)` shows exactly those (not the other citations among the same papers). `scatterLayout(year, generation)` lays it out as a timeline: year across, citation depth up. Here the path runs from Chiba et al. (1985) and Tamassia et al. (1988) through Kamada–Kawai (1989) and Fruchterman–Reingold (1991) to the multilevel methods of 2000–2006 (Walshaw, Harel and Koren, Hachul and Jünger). It then turns to edge bundling (2008–2010, including Holten's force-directed bundling), then to the visualisation of dynamic graphs (2011–2016), and ends at a 2024 review of how layout algorithms are evaluated. That's the field's arc from drawing small graphs to drawing large and changing ones.

Variations. `sizeBy(spc, 1, 10)` (before `showOnly`) widens edges by their path count, showing the main path and its strongest tributaries.

Pitfalls. The main path is only as good as the paper set. It can't pass through work that the search missed. Citation cycles (papers citing each other's preprints) are ignored, so check `feedbackEdges()` if the path looks odd.

Name the research fronts Intermediate

You want the field's subareas, each with a name.

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
leidenClusters()
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
for grp in groupBy(leiden_class)[:6]:
    best = enrichment(grp, kw, show=False)[0]
    print(len(grp), best["set"], sortBy(grp, indegree)[-1].label)
```

How it works. Communities in a citation network are groups of papers that cite each other: research fronts. `enrichment` names each one by its most over-represented keyword (*Biology: Find modules and name them*), and the most cited member makes a good exemplar. The fronts found here are force-directed layout (around Fruchterman–Reingold), node-link diagrams and surveys (around Di Battista et al.'s book), planar and linear-time algorithms, aesthetics and symmetry (around Purchase's readability studies), hierarchical and Graphviz-style layout (around Gansner et al.), and dynamic network visualisation.

Co-authorship and keywords

Who writes with whom? Intermediate

You have a table of papers and their authors. **You want** the collaboration network.

Ingredients: `vis-paper-authors.csv` · `makeFromTwoMode` · `project`

```
makeFromTwoMode("vis-paper-authors.csv", "paper", "author")
crowded = [n for n in g.nodes if n.kind == "paper" and n.degree >= 20]
len(crowded)
for p in crowded: remove(p)
```

```
project(kind, "author")
removeDisconnected()
for n in sortBy(affiliations)[-5:]: print(n.label, n.affiliations, n.degree)
```

How it works. `makeFromTwoMode(file, "paper", "author")` reads one row per authorship as a two-mode network: paper and author nodes, with `kind` saying which. `project(kind, "author")` then links authors who wrote together (*SNA: Project onto one kind*). `affiliations` is each author's number of papers in the set. Peter Eades, Stephen Kobourov and Roberto Tamassia have the most.

Pitfalls. One paper with 25 authors adds 300 ties to the projection, and swamps the collaboration structure. Here, four such papers turn out to be atmospheric science: the title search for "force-directed" matched "direct … forcing". Look at the outliers before projecting. They're where data problems hide. Removing papers with 20 or more authors fixes both problems.

Map the keywords Intermediate

```
makeFromTwoMode("vis-paper-keywords.csv", "paper", "keyword")
project(kind, "keyword", method="jaccard")
remove(affiliations < 8)
remove(weight < 0.05)
showOnly(giantComponent())
louvainClusters()
colorize(louvain_class)
sizeBy(affiliations, 12, 60)
labelBy(label)
labelSizeBy(affiliations)
forceAtlas2Layout()
```

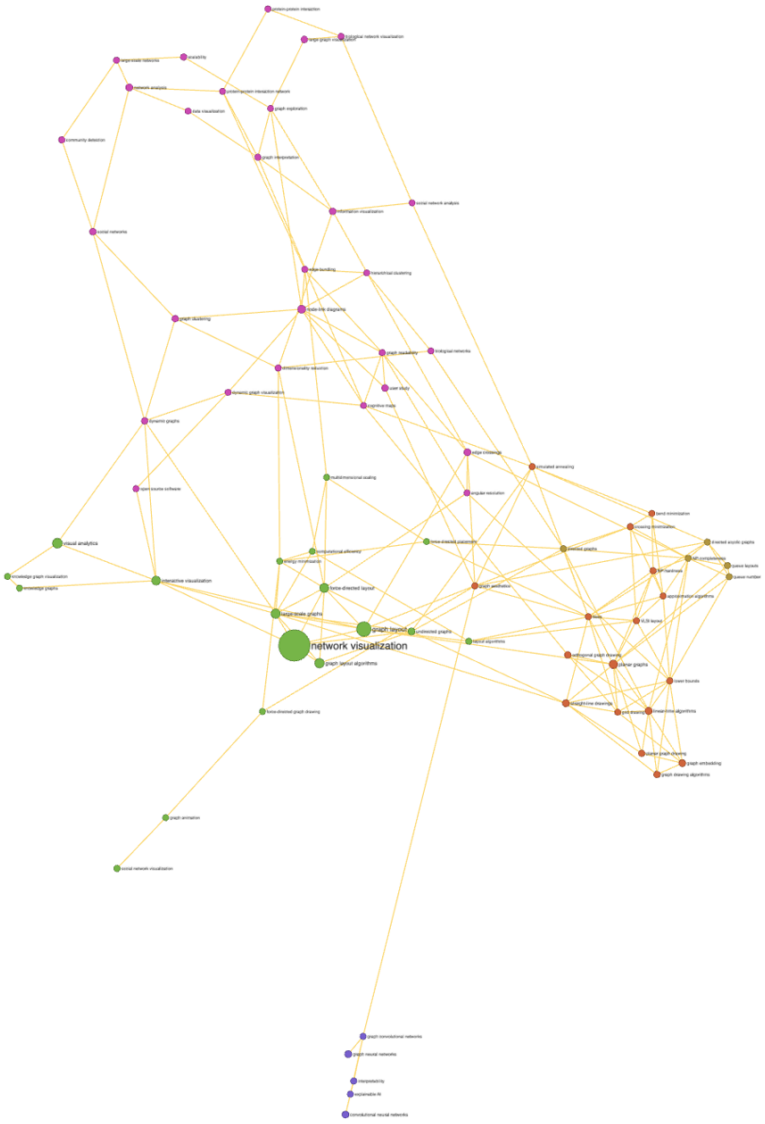


Figure 4: Keyword co-occurrence in graph-visualisation papers

How it works. Projecting papers × keywords onto the keywords links two keywords when papers share them. Jaccard weights (shared papers ÷ papers with either) keep a ubiquitous keyword like "network visualization" (382 papers) from linking to everything. Removing rare keywords (in fewer than 8 papers) and weak links leaves 86 keywords. The 73 that stay connected form a thematic map whose communities are the field's topics. Filter first, then keep the largest component, since each removal can strand more nodes.

Similarity

Link documents with similar content Intermediate

You have documents described by features (keywords, term counts, measurements). **You want** each document linked to the ones most like it, whether or not they cite each other.

Ingredients: `vis-paper-keywords.csv`, `vis-papers-nodes.csv` · `makeFromSimilarity`

```
makeFromSimilarity("vis-paper-keywords.csv", features="keyword", values="score",
↪  k=4)
makeFromCSV("vis-papers-nodes.csv")
leidenClusters()
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
for grp in groupBy(leiden_class)[:5]:
    print(len(grp), enrichment(grp, kw, show=False)[0]["set"])
```

How it works. `makeFromSimilarity` turns a table into a network: each row (here, each paper) is linked to its `k` most similar rows. The table can be *wide* (one column per feature) or, as here, *long*, with one row per paper, keyword and score (`features=` and `values=` name the columns). `method` chooses the similarity:

method	similar when	use for
`"cosine"` (default)	profiles point the same way	term weights, keyword scores
`"correlation"`	values rise and fall together	expression profiles, time series
`"jaccard"`	the same features are present	yes/no data
`"euclidean"`	values are close	measurements on one scale

`threshold=0.5` links every pair at least that similar instead of the top `k`, and `mutual=true` keeps only pairs that are in each other's top `k`. `makeFromCSV` then adds titles and years from the node table.

Pitfalls. A k-nearest-neighbour network links every node to *k* others, even an outlier with no real neighbours, so look at the weights (`plotHistogram(weight)`). A threshold network leaves outliers isolated, which is often more honest.

Live sources

Wikipedia's link neighbourhood Basic

```
makeFromWikipedia("Graph drawing", limit=100)
sizeBy(indegree, 4, 30)
labelBy(label)
forceAtlas2Layout()
```

How it works. `makeFromWikipedia(title)` opens an article, the articles it links to (up to `limit`), and every link among them. Sizing by `indegree` shows which linked articles the neighbourhood itself points to most: the topic's core concepts. Pages that citation templates link to everywhere

(such as “Doi (identifier)”) are left out. `lang=` picks another language edition. Comparing editions is a study in itself.

The bow tie of a link network Intermediate

You want to know how a directed network of links hangs together: which pages can reach which, and which only receive or only send.

Ingredients: `polblogs-sample.gdf` · `bowTie` · `bowtie`

```
makeFromFile("polblogs-sample.gdf")
bowTie()
colorize(bowtie)
legendFor(bowtie)
```

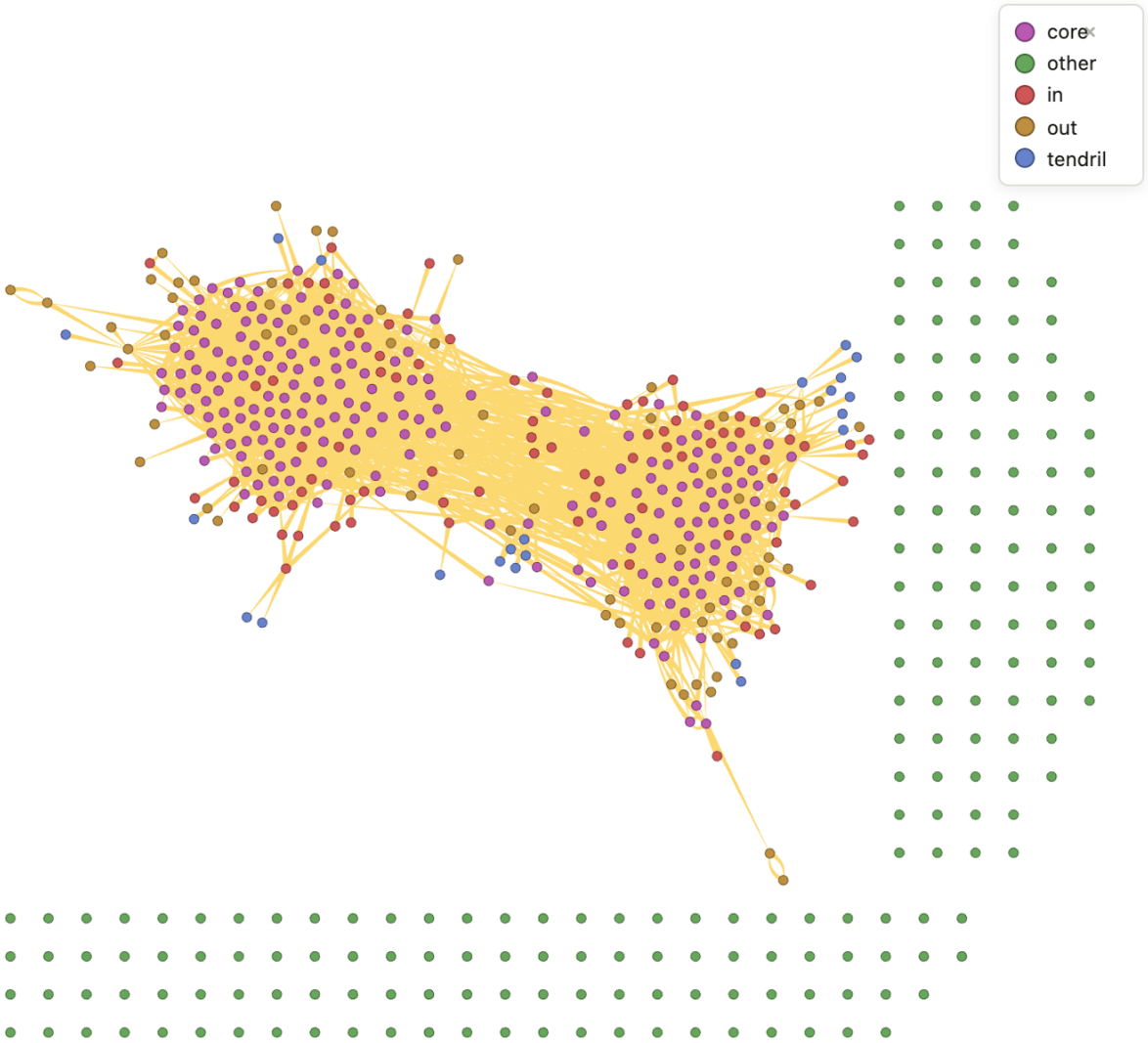


Figure 5: The bow tie of 600 political blogs

How it works. In 2000 Broder and colleagues found that the Web is shaped like a bow tie. A strongly connected *core* of pages can all reach one another. An *in* set can reach the core but not be reached from it, and an *out* set is reached from the core but doesn't link back. *Tendrils* hang off in or out, *tubes* run from in to out past the core, and the rest isn't connected to the core at all. `bowTie()` sorts the nodes into those parts, stores the part in the field `bowtie`, and returns the sizes:

part	blogs	share	the Web, 1999
core	224	37%	28%
in	89	15%	21%

part	blogs	share	the Web, 1999
out	69	12%	21%
tendrils	22	4%	22% (with tubes)
outside	196	33%	8%

The blogs' core is liberal and conservative alike (92 of 224 liberal), so the two camps are linked into one structure even if they link mostly to their own side. Many blogs in `in` link to the big blogs but nobody links back to them.

Pitfalls. This is a random sample of 600 of the 1,490 blogs, which keeps the proportions but cuts links: the third outside the bow tie is partly an effect of sampling. In-sets and out-sets depend on direction, so check that edges run from the linking page to the linked one.

Any field's literature, from OpenAlex Basic

```
makeFromOpenAlex("edge bundling", limit=200)
removeDisconnected()
mp = mainPath()
leidenClusters()
```

How it works. `makeFromOpenAlex(words)` builds the same kind of citation network as `vis-papers.gdf` for any topic: the most cited works with the words in their titles, and the citations among them. `mode="authors"` gives the paper–author network for co-authorship recipes. Every recipe in the citation chapters then works on the field you care about. Remember the pitfalls of title search: synonyms are missed and homonyms are caught.

Knowledge graphs

Query Wikidata Intermediate

You want a network from a public knowledge graph, here "influenced by" links among philosophers.

```
q = """SELECT ?source ?sourceLabel ?target ?targetLabel WHERE {
  ?source wdt:P106 wd:Q4964182; wdt:P737 ?target .
  ?target wdt:P106 wd:Q4964182 .
  SERVICE wikibase:label { bd:serviceParam wikibase:language "en". }
} LIMIT 400"""
makeFromSPARQL(q)
sizeBy(indegree, 6, 40)
labelBy(label)
g.nodes.labelpos = "outside"
forceAtlas2Layout()
```

How it works. `makeFromSPARQL(query)` runs a SPARQL query (on Wikidata unless you give another endpoint) whose rows bind `?source` and `?target`. `?sourceLabel`/`?targetLabel` become labels, and any other variables become edge fields. In Wikidata, `P106` is occupation, `Q4964182` is philosopher and `P737` is "influenced by", so edges run from each philosopher to their influences. The most cited influences are Aristotle, Marx, Plato, Hegel and Descartes.

Pitfalls. Knowledge graphs are incomplete in uneven ways: famous, Western and recent subjects are better covered. `LIMIT` also truncates silently. Read the result as "what Wikidata records", and check the total count with a `COUNT(*)` query.

History and geography

Where was a field made, and when? Intermediate

You want to see the geography of a body of work change over time: which places took part, and when they joined.

Ingredients: `vis-institutions.gdf` · `geoLayout` · `showTime`

```
makeFromGDF("vis-institutions.gdf")
geoLayout(projection="equirectangular")
sizeBy(papers, 3, 24)
colorBy(firstyear, blue, red)
legendFor(firstyear)
g.edges.color = "120,120,120,30"
showTime(1999, firstyear)
```

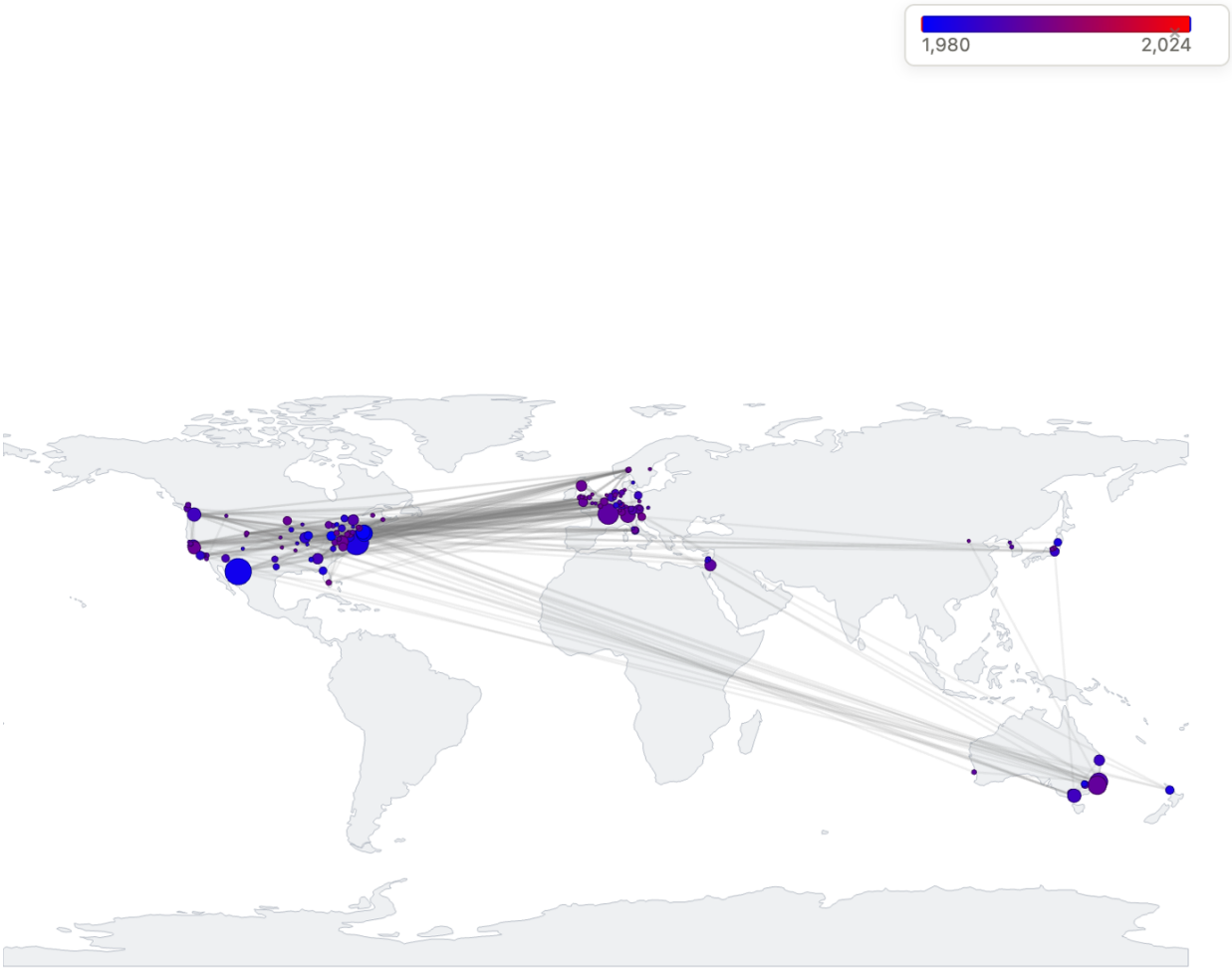


Figure 6: Institutions publishing on graph drawing before 2000

How it works. Each institution sits where it is (`geoLayout`), sized by its number of papers and coloured by the year it first appears. `showTime(1999, firstyear)` shows only the institutions active by 1999, and the edges between them. Step the year and the map fills in. *View* → *Timeline* gives a slider. In the 1980s the field is mostly American (20 of 32 institutions). Germany, Britain, Australia and Canada join in the 1990s, and by the 2010s France has the second-largest group. In the 2020s the largest group of newly active institutions is Chinese (42 of 143).

Pitfalls. A paper set built from title searches in English, sorted by citations, under-counts recent work (it hasn't been cited yet) and work published in other languages. The off-topic atmospheric-

science papers from *Who writes with whom?* bring their institutions too. A map makes a sample look like a census. Say how the sample was drawn.

Do collaborations cross borders? Intermediate

```
makeFromGDF("vis-institutions.gdf")
for d in (1990, 2000, 2010, 2020):
    ties = [e for e in g.edges if d <= e.firstyear < d + 10]
    abroad = [e for e in ties if e.node1.country != e.node2.country]
    print(d, len(ties), round(len(abroad) / len(ties), 2))
geoLayout(projection="equirectangular")
g.edges.visible = false
for e in g.edges: e.visible = e.node1.country != e.node2.country
bundleEdges()
```

How it works. Each edge is a collaboration between two institutions, dated by their first joint paper. Comparing the two ends' countries shows how international the field's collaborations are: about half the ties formed in each decade from the 1990s to the 2010s join institutions in different countries, rising to 64% in the 2020s. The last lines show only the cross-border ties on the map, bundled so the transatlantic and transpacific corridors stand out.

Case studies

Case study: the contest in "A Scandal in Bohemia" Basic

```
cast = {"Holmes": ["Holmes", "Sherlock"],
        "Irene Adler": ["Irene Adler", "Irene", "Miss Adler", "the woman",
                        "Mrs. Norton"],
        "the King": ["King", "Majesty", "Count Von Kramm", "Von Kramm"],
        "Godfrey Norton": ["Godfrey Norton", "Godfrey", "Mr. Norton"]}
makeFromText("scandal-in-bohemia.txt", cast)
[(n.label, n.first, n.units) for n in sortBy(first)]
```

The `first` field gives the order of entrances. Irene Adler is named in the third paragraph (Watson's frame: she is "the woman"), long before the King appears (paragraph 47), and Godfrey Norton comes last. Holmes is tied to both the King and Irene Adler, and Norton only to her. The King and Irene Adler, whose quarrel is the story, are never named in the same paragraph, though he spends pages telling her story. In dialogue, speakers are rarely named, so the paragraphs where he speaks about her don't mention him. A co-mention network shows who is *named* together. Here that means the plot runs through Holmes, but conversations don't show up. For a play, where every speech is labelled with its speaker, link speakers by scene instead.

Case study: how graph drawing grew up Advanced

```
makeFromGDF("vis-papers.gdf")
removeDisconnected()
```

```
leidenClusters()
mp = mainPath()
addNodeField("decade", INTEGER, 0)
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
names = {}
for grp in groupBy(leiden_class)[:8]:
    names[grp[0].leiden_class] = enrichment(grp, kw, show=False)[0]["set"]
for n in g.nodes: n.decade = n.year // 10 * 10
for d in (1980, 1990, 2000, 2010):
    fronts = sorted(groupBy(decade == d, leiden_class), key=len)[-3:]
    top = [(names.get(f[0].leiden_class, "other"), len(f)) for f in
↪   reversed(fronts)]
    print(d, len(decade == d), top)
```

This puts the pieces together. Each research front is named by its top keyword, and the papers of each decade are counted by front. In the 1980s and 1990s the largest front is algorithmic ("linear-time algorithms": planar and orthogonal drawing), with layout second. By the 2000s, the force-directed "network visualization" front and the "node-link diagrams" front have caught up. In the 2010s node-link work (bundling, surveys, evaluation) leads. Overlay the main path (`mp.color = red`), which moves through the same fronts in the same order. One citation network and three recipes (communities, their names, the main path) give a defensible short history of a field. Check it against what its practitioners say, of course.

Capstone

Capstone: a literature explorer Advanced

The problem. A newcomer to a field wants to explore its literature as a living thing: how it grew, what its fronts are, which papers matter, and how the main line of work runs through it. This recipe builds a literature explorer on the graph-drawing papers:

- a year slider grows the citation network up to the chosen year;
- a drop-down highlights one research front, named by keyword;
- a checkbox overlays the main path;
- a click on a paper shows its title, venue and citation count, and a right-click shows everything that builds on it;
- the legend explains the colours by front.

```
makeFromGDF("vis-papers.gdf")
_ = removeDisconnected()
forceAtlas2Layout()
_ = leidenClusters()
kw = readSets("vis-paper-keywords.csv", item="paper", set="keyword")
fronts = {}
for grp in groupBy(leiden_class)[:8]:
    fronts[enrichment(grp, kw, show=False)[0]["set"]] = grp[0].leiden_class
colorize(leiden_class)
mp = mainPath()
```

```
sizeBy(indegree, 3, 28)

def until(y):
    showTime(int(y), year)
    shown.text = f"{len(year <= int(y))} papers by {int(y)}"

def front(name):
    g.nodes.labelvisible = false
    members = (leiden_class == fronts[name])
    for n in sortBy(members, indegree)[-6:]: n.labelvisible = true
    center(members)

def main_path(on):
    g.edges.width = 1
    if on:
        for e in mp:
            if e in g.edges: e.width = 6

def paper(n):
    about.text = f"{n.label}: {n.title} ({n.venue or 'no venue'}); cited
 ↪  {n.cites} times overall"

shown, about = Label(""), Label("click a paper")
ui.dock(Panel("Literature", Slider(1980, 2024, 2024, label="year",
 ↪  onChange=until),
             Dropdown(sorted(fronts), label="research front", onChange=front),
             Checkbox("main path", onChange=main_path), shown, about,
 ↪  position="bottom"))
graphevents.clickNode = paper
item = NodeEditorPopup.addItem("Show what builds on this")
item.menuEvent = lambda n: showOnly(ancestors(n) | [n])
until(2024)
center(giantComponent())
```

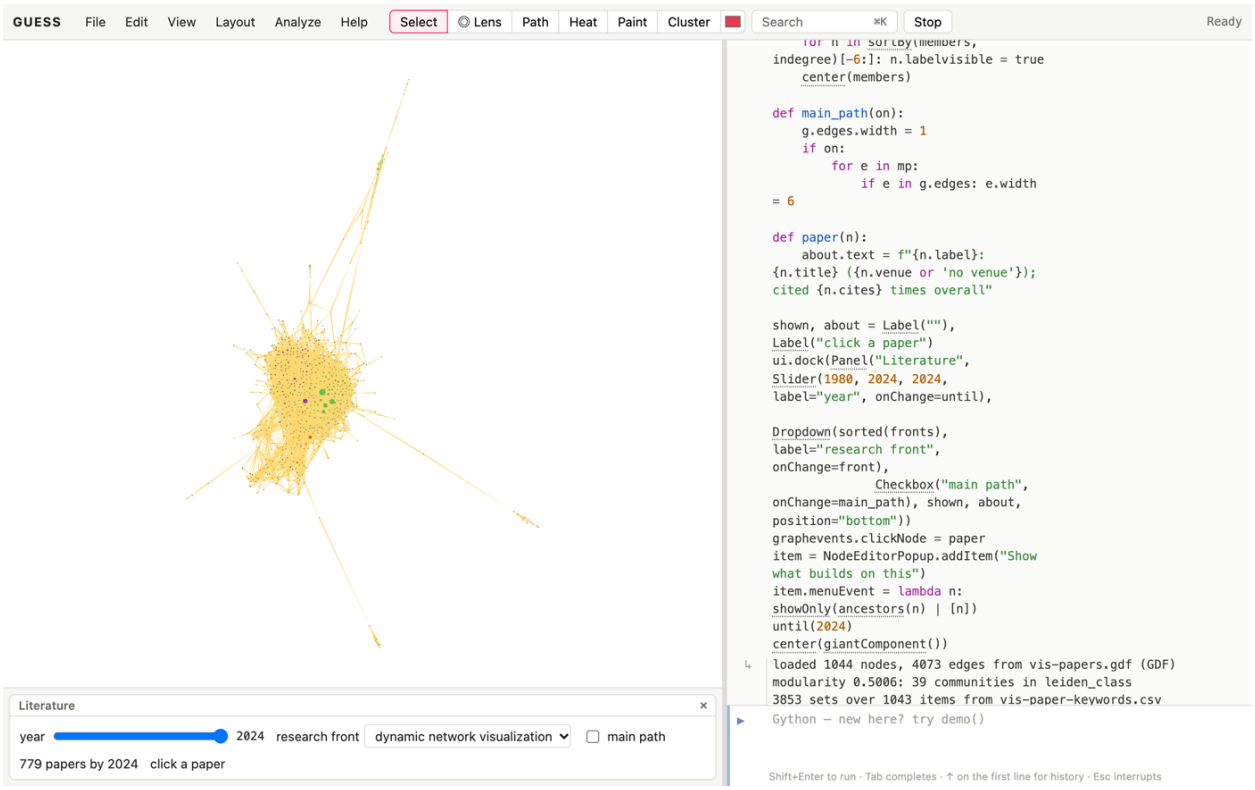


Figure 7: A literature explorer: graph-drawing research, coloured by research front

How it works.

- **The slider.** `showTime(t, year)` shows the papers published by year `t` and the citations among them, so dragging it replays the field's growth.
- **The fronts.** They're Leiden communities named by their top keyword (the research-fronts recipe), and the drop-down labels each front's most cited papers.
- **The main path.** `mainPath()` is computed once. The checkbox widens its edges.
- **The right-click item.** It uses `ancestors`: in a citation network (citing → cited), the ancestors of a paper are everything that cites it, directly or through others, which is its intellectual descendants.